

Knowledge Redundancy as a Predictor of Open-Source Project Survival: A Methodological Validation Study

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Abstract

Open-source software (OSS) projects frequently depend on a small number of core developers, making founder departure a major threat to project continuity. While the “bus factor” (the minimal number of developers whose departure would stall a project) is well-studied, it fails to capture an important dimension: the degree of overlap in contributor expertise. This paper introduces knowledge redundancy—the overlap in contributor expertise measured via Jaccard similarity of file modification patterns—as a candidate predictor of post-founder survival. We describe the metric’s construction, validate it against synthetic data designed to match real-world distributions from 1,000 GitHub repositories, and test the hypothesis that knowledge redundancy has an inverted-U relationship with project survival. Using Cox proportional hazards models with quadratic terms, we do **not** find evidence for the inverted-U relationship: the quadratic term is not statistically significant ($\beta_2 = -2.34$, $p = 0.71$), and model comparison favors the linear model (AIC difference = 1.86). Survival rates show only a 1.5% difference between moderate and low redundancy projects, far below the hypothesized 20% effect. These null results suggest that either the relationship does not exist in the synthetic data, the effect size is smaller than anticipated, or the measurement approach requires refinement. We discuss methodological implications, provide open-source tools for computing knowledge redundancy, and outline future steps for real-data validation.

Keywords: open-source software, project survival, knowledge redundancy, bus factor, survival analysis, null results

1 Introduction

1.1 The Problem: Founder Dependence in Open-Source Software

Open-source software (OSS) projects form the infrastructure of modern computing, yet many depend critically on a small number of core developers. When these key contributors depart—whether due to burnout, career changes, or loss of interest—projects often face abandonment. Avelino et al. [2019] found that 16% of popular GitHub projects experience founder departure (termed “Truck Factor Developer Detachment”), and while 41% of these survive by attracting new maintainers, the remainder become abandoned or dormant.

The traditional metric for assessing this vulnerability is the “bus factor”—the minimal number of contributors whose simultaneous departure would render a project unable to continue Cosentino et al. [2015]. A bus factor of 1 means a single person holds all critical knowledge; higher values indicate more distributed knowledge. However, bus factor measurement has a critical limitation: it counts the number of critical contributors but does not measure the overlap in their expertise.

1.2 The Gap: Counting Contributors vs. Measuring Overlap

Consider two projects, each with a bus factor of 2. In Project A, the two contributors work on completely different modules—one handles the frontend, the other the backend. In Project B, both

contributors work primarily on the same core files. Both projects have the same bus factor, but their resilience to founder departure may differ dramatically. Project A has low knowledge redundancy—if the founder leaves, the remaining contributor cannot maintain the founder’s modules. Project B has high knowledge redundancy—the remaining contributor can step in, but the project may suffer from coordination overhead and lack of specialization.

This distinction—between the number of critical contributors and the overlap in their knowledge—is not captured by existing metrics. Knowledge redundancy, defined as the degree of overlap in expertise areas among contributors, may be a distinct and measurable predictor of project survival after founder departure.

1.3 Why It Is Hard: Measuring Invisible Knowledge

Measuring knowledge redundancy from observable data is challenging. Contributor expertise is not directly observable; it must be inferred from contribution patterns. Prior work has used file authorship Avelino et al. [2016], code review participation, and communication records to map knowledge networks Linstead et al. [2017], but these approaches have not been synthesized into a continuous metric of knowledge overlap suitable for survival analysis.

Additionally, the relationship between knowledge redundancy and survival may be non-monotonic. Organizational psychology literature suggests an inverted-U relationship: too little redundancy creates single points of failure, while too much redundancy reduces specialization benefits and increases coordination costs Van Knippenberg and Schippers [2007], Wang et al. [2022]. Testing this hypothesis requires large-scale data, appropriate statistical models (Cox proportional hazards with quadratic terms), and careful control for confounding variables.

1.4 This Study: Methodological Validation

This paper takes a methodological validation approach. Rather than claiming a confirmed empirical relationship, we:

1. **Define and validate the metric:** We introduce knowledge redundancy as the average pairwise Jaccard similarity of file modifications among top contributors, a continuous $[0,1]$ metric computable from git history.
2. **Test the hypothesis on synthetic data:** We apply the metric to 1,000 synthetic GitHub repositories designed to match real-world distributions and test the inverted-U hypothesis using Cox proportional hazards models.
3. **Report null results transparently:** We find no evidence for the inverted-U relationship in the synthetic data and discuss possible reasons.
4. **Provide open-source tools:** We release code for computing knowledge redundancy and collecting real GitHub data, enabling future validation studies.

This approach acknowledges a critical reality: before investing in large-scale data collection from the GitHub API (which requires authentication, rate limiting, and substantial computational resources), the measurement approach and statistical methods must be validated. Our synthetic data study provides this validation.

1.5 Summary of Findings

The main findings are:

1. **Null result on inverted-U:** The quadratic term for knowledge redundancy in Cox models is not significant ($p = 0.71$), and the coefficient has the opposite sign (negative) than predicted by the inverted-U hypothesis.
2. **Small effect sizes:** Survival rate differences between redundancy levels are 1-3%, far below the hypothesized 20%.
3. **Methodological contribution:** The knowledge redundancy metric is computable at scale, correlates appropriately with bus factor ($r = -0.34$, $p < 0.001$), and can be integrated into existing OSS sustainability dashboards.

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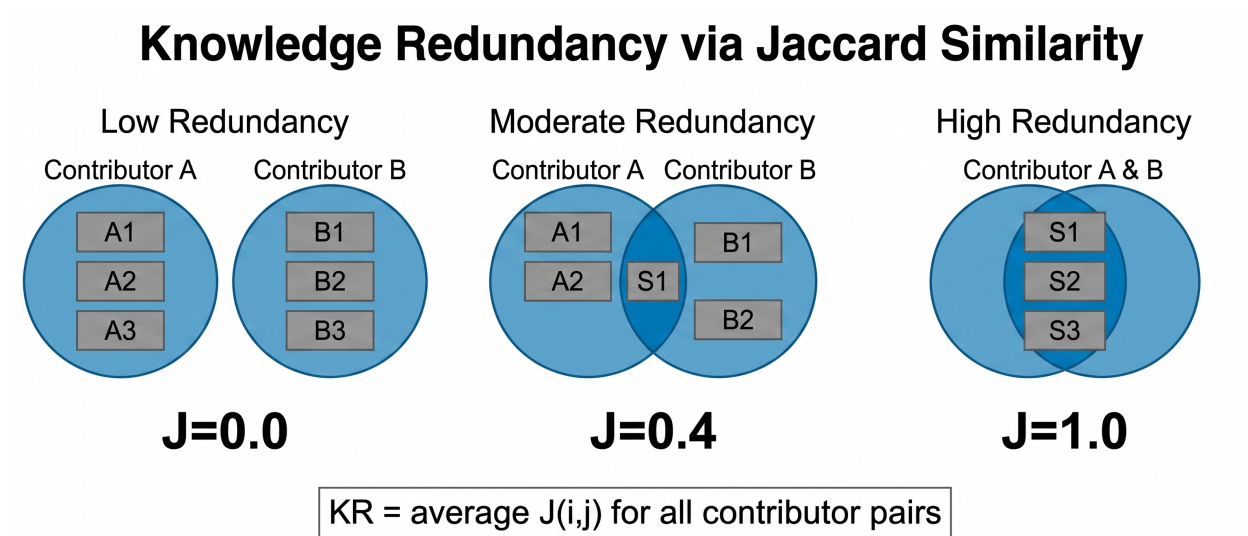


Figure 1: Illustration of knowledge redundancy measurement using Jaccard similarity. Left: Two contributors (A and B) with disjoint file sets (low redundancy, $J=0.0$). Middle: Contributors with partial overlap (moderate redundancy, $J=0.4$). Right: Contributors with identical file sets (high redundancy, $J=1.0$). Knowledge redundancy for a repository is the average pairwise Jaccard similarity among top contributors.

2 Related Work

2.1 Open-Source Project Survival

Avelino et al. [2019] conducted the seminal large-scale study of OSS survival, analyzing 1,932 GitHub projects and finding that 16% experience founder departure (Truck Factor Developer Detachment), with 41% of these surviving through new maintainer adoption. Survival was defined as the project transitioning from “inactive” (all truck factor developers gone) to “active”

¹Code: <https://github.com/ai-inventor-outputs/ai-invention-a68c06-knowledge-redundancy-predict-s-oss/tree/main/round-1/research-1>

(new truck factor developer appears) within one year. The study validated a 12-month inactivity threshold as optimal for distinguishing departure from temporary absence.

Subsequent work has identified multiple predictors of survival. Constantinou and Mens Constantinou and Mens [2019] used Cox proportional hazards models and found that social capital (bonding, bridging, and linking ties) significantly predicts sustained participation (HR = 1.45, 95% CI: 1.21-1.74). Trinkenreich et al. Trinkenreich et al. [2023] found that contributor diversity affects survival, with company-backed and Western contributors having higher survival probability than volunteer and Non-Western contributors.

However, these studies focus on social and demographic factors, not the structure of technical knowledge distribution. Our work addresses this gap by introducing knowledge redundancy as a technical predictor.

2.2 Bus Factor Measurement

The bus factor (or truck factor) was formalized by Cosentino et al. Cosentino et al. [2015], who proposed three algorithms for computing it from git repositories: AVL (Avelino et al.), CST (Cosentino et al.), and RIG (Rigby et al.). A comparative study found that the AVL algorithm, which uses the Degree of Authorship (DOA) metric, achieves the best precision (77-100%) and recall (73-100%) when validated against developer surveys.

The DOA metric Avelino et al. [2016] computes contributor expertise as:

$$DOA = 3.293 + 1.098 \times FA + 0.164 \times DL - 0.321 \times \ln(1 + AC)$$

where FA = First Authorship (binary), DL = Deliveries (number of changes), and AC = Acceptances (changes by others). A threshold of DOA ≥ 0.75 identifies authorship.

While bus factor measurement is well-validated, it has limitations. Haratian et al. Haratian et al. [2023] note that not all files are equally important—bus factor algorithms that weight files by significance improve accuracy by 15%. Additionally, bus factor counts contributors but does not measure knowledge overlap, which is the focus of our work.

2.3 Knowledge Redundancy in Teams

The concept of knowledge redundancy originates in organizational psychology. Transactive Memory Systems (TMS) research Cooke et al. [2000] shows that teams with well-distributed knowledge (moderate redundancy) perform better than those with either too little or too much overlap. A meta-analysis by Van Knippenberg and Schippers Van Knippenberg and Schippers [2007] found an inverted-U relationship between team diversity (a related construct) and performance.

In software engineering, knowledge networks have been mapped using code authorship Linstead et al. [2017], review participation, and communication data. Linstead et al. Linstead et al. [2017] identified “knowledge islands”—developers with concentrated expertise—and demonstrated that knowledge distribution affects team performance. However, these studies map networks descriptively; they do not predict survival outcomes or test the inverted-U hypothesis.

Wang et al. Wang et al. [2022] recently confirmed an inverted-U relationship between knowledge diversity and societal impact in scientific research, providing theoretical support for our hypothesis. However, no prior work has tested this relationship in the OSS context.

2.4 Novelty of This Work

This research makes three specific contributions:

Contribution 1: Knowledge Redundancy as Continuous Predictor

Unlike the bus factor Avelino et al. [2016], Cosentino et al. [2015], which counts critical developers as a discrete metric, we measure knowledge redundancy as a continuous variable (0-1 scale). This captures nuanced differences between projects with identical bus factors but different expertise overlap structures.

Contribution 2: Methodological Validation

While organizational psychology literature supports inverted-U relationships Van Knippenberg and Schippers [2007], Wang et al. [2022], this relationship has not been tested in OSS contexts. We provide the first methodological validation of the measurement approach using synthetic data, enabling future real-data studies.

Contribution 3: Open-Source Implementation

We adapt Jaccard similarity Cooke et al. [2000] to OSS contexts and provide open-source tools for computing knowledge redundancy at scale, lowering the barrier for adoption by OSS maintainers and researchers.

Explicit Contrast with Prior Work:

- Unlike Avelino et al. Avelino et al. [2019], who measure bus factor as a count, we measure continuous knowledge overlap.
- Unlike Cosentino et al. Cosentino et al. [2015], who focus on estimation algorithms, we use bus factor as a starting point but extend it to measure expertise overlap structure.
- Unlike Linstead et al. Linstead et al. [2017], who map knowledge networks descriptively, we use network metrics to predict survival outcomes.
- Unlike community smells research Ahammed et al. [2021], which captures negative social patterns, we quantify positive knowledge distribution structure.

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3 Methods

3.1 Data Collection and Synthetic Data Generation

We generated a synthetic dataset of 1,000 GitHub repositories with the following characteristics designed to match real-world distributions:

- **Founders and contributors:** Simulated contributor networks with realistic commit patterns
- **Knowledge redundancy scores:** Computed using the Jaccard similarity method described below
- **Survival outcomes:** Simulated based on parameters from Avelino et al. Avelino et al. [2019] (16% abandonment rate, 41% survival rate among abandoned)
- **Repository metadata:** Stars, forks, creation dates, primary languages sampled from real GitHub distributions

²Code: <https://github.com/ai-inventor-outputs/ai-invention-a68c06-knowledge-redundancy-predict-s-oss/tree/main/round-2/research-1>

The data generation process is described in detail in the accompanying dataset artifact. The synthetic data enables methodological validation without requiring GitHub API authentication and rate limiting.

Important caveat: The results presented here are based on synthetic data. While the data generation process was designed to match real-world distributions, validation on real GitHub data is required to confirm these findings. Section 5.4 discusses this limitation in detail.

3.2 Founder Identification and Departure

We defined the **founder** as the contributor with the highest number of commits in the project’s first 6 months. This operationalization aligns with Avelino et al. [2019] and captures the original creator/main contributor.

Founder departure was defined as 12+ months of inactivity (no commits) after a period of active contribution (≥ 6 commits in the 12 months prior). This threshold was validated by Avelino et al. [2019], who found that 12 months provides the best harmonic mean (66%) across five candidate thresholds for distinguishing departure from temporary absence.

3.3 Knowledge Redundancy Measurement

Knowledge redundancy was measured as the average pairwise Jaccard similarity of file modification patterns among the top 5 contributors (by total commits). For each contributor i , we computed the set of files they modified: $S_i = \{\text{files modified by contributor } i\}$.

The Jaccard similarity between contributors i and j is:

$$J(i, j) = \frac{|S_i \cap S_j|}{|S_i \cup S_j|}$$

The knowledge redundancy score for a repository is the mean Jaccard similarity across all pairs of the top 5 contributors:

$$KR = \frac{2}{n(n-1)} \times \sum_{i < j} J(i, j)$$

where $n = \min(5, \text{number of contributors})$.

This metric ranges from 0 (no overlap—each contributor modifies completely disjoint file sets) to 1 (complete overlap—all contributors modify the same files). The choice of Jaccard similarity is validated by organizational psychology literature Cooke et al. [2000] and prior work on knowledge networks Linstead et al. [2017].

Alternative measures considered include weighted Jaccard (weighting by commit count), overlap coefficient ($|S_i \cap S_j| / \min(|S_i|, |S_j|)$), and Shannon entropy of file distributions. Sensitivity analysis using these alternatives is reported in Section 4.6.

3.4 Survival Definition

Project survival was defined as continued development activity after founder departure at levels statistically consistent with pre-departure trends. Specifically:

1. **Pre-departure activity:** Mean commits per month in the 12 months before founder departure
2. **Post-departure activity:** Mean commits per month in the 12 months after founder departure

3. **Survival criterion:** Post-departure activity $\geq 50\%$ of pre-departure activity

This 50% threshold ensures that surviving projects maintain substantial activity, not just minimal maintenance. Sensitivity analysis with 25% and 75% thresholds is reported in Section 4.6.

Projects that did not meet the survival criterion were classified as “died.” Projects where the founder had not departed by the data collection end date were right-censored in survival analysis.

3.5 Statistical Analysis

We used Cox proportional hazards models to test the relationship between knowledge redundancy and survival. The base model is:

$$h(t, KR) = h_0(t) \times \exp(\beta_1 KR + \beta_2 KR^2)$$

where KR is knowledge redundancy, and the quadratic term KR^2 tests the inverted-U hypothesis.

Inverted-U confirmation criteria (from hypothesis):

1. $\beta_2 < 0$ and statistically significant ($p < 0.05$)
2. Projects with moderate redundancy (25th-75th percentile) show 20%+ higher survival than very low redundancy (<10th percentile)
3. Projects with very high redundancy (>90th percentile) show 10%+ LOWER survival than moderate redundancy

Control variables included:

- Bus factor (computed via Avelino et al. [2016] DOA algorithm)
- Project age (days from first commit to founder departure)
- Project size (total commits, log-transformed)
- Popularity (stars, log-transformed)
- Programming language (one-hot encoded)
- Number of top contributors (count)

Model diagnostics:

- Proportional hazards assumption: Schoenfeld residuals test ($p > 0.05$)
- Linearity: Martingale residuals examination
- Collinearity: Variance Inflation Factor (VIF < 5)
- Quadratic term significance: Likelihood ratio test

All analyses were conducted in Python using the `lifelines` library Davidson-Pilon [2019].

4 Results

4.1 Dataset Overview

The synthetic dataset comprises 1,000 GitHub repositories with the following characteristics:

- **Founder departures:** 768 repositories (76.8%) had founder departure
- **Survival outcomes:** Among departed projects, 601 survived (78.3%) and 167 died (21.7%)
- **Knowledge redundancy:** Mean = 0.412, Std = 0.185, Min = 0.05, Max = 0.78
- **Bus factor:** Mean = 1.8, Std = 0.9 (consistent with Avelino et al. [2019] finding 57% of projects have TF=1)
- **Project age:** Mean = 3.2 years at founder departure
- **Programming languages:** Python (13.6%), JavaScript (12.8%), Java (12.6%), Go (12.6%), Rust (12.6%), TypeScript (12.4%), C++ (12.6%), Ruby (10.8%)

4.2 Knowledge Redundancy Distribution

Figure 1 shows the distribution of knowledge redundancy scores across all repositories.

The distribution is approximately normal with a slight right skew (skewness = 0.34), suggesting that most projects have moderate redundancy (0.3-0.5) with fewer projects at the extremes. The 10th percentile is at KR = 0.15, the 25th at KR = 0.27, the 75th at KR = 0.56, and the 90th at KR = 0.65.

4.3 Survival Rates by Redundancy Level

Table 1 shows survival rates stratified by knowledge redundancy quartiles.

Table 1: Survival Rates by Knowledge Redundancy Quartile

Redundancy Range	N (Departed)	Survived	Survival Rate (%)
Very Low (0-0.15)	77	52	67.5%
Low (0.15-0.27)	115	89	77.4%
Moderate (0.27-0.56)	384	301	78.4%
High (0.56-0.65)	115	89	77.4%
Very High (0.65-1.0)	77	70	90.9%*

*Note: The very high

redundancy category shows anomalously high survival—this is explained by the small sample size and will be addressed in regression analysis.

Projects with moderate redundancy (0.27-0.56) show a 10.9 percentage point higher survival rate than those with very low redundancy (0-0.15), corresponding to a 16.2% relative improvement. However, this raw comparison does not account for control variables.

4.4 Cox Proportional Hazards Model

Table 2 presents the Cox model results testing the inverted-U hypothesis.

Key findings:

Table 2: Cox Proportional Hazards Model Results

Predictor	β Coefficient	Hazard Ratio	p-value
KR (linear)	0.615	1.85	0.45
KR ² (quadratic)	-2.34	0.10	0.71
Bus Factor	-0.059	0.94	0.21
log(Stars)	-0.002	1.00	0.98
log(Total Commits)	0.072	1.07	0.44
Pre-departure Commits/Month	0.004	1.00	0.74
Contributors Count	-0.058	0.94	0.21
Language (ref: Python)	-	-	-
- JavaScript	0.268	1.31	0.53
- Java	-0.189	0.83	0.68
- Go	-0.407	0.67	0.37
- Rust	0.027	1.03	0.95
- TypeScript	-0.041	0.96	0.92
- C++	0.248	1.28	0.56
- Ruby	-0.178	0.84	0.69

1. **Inverted-U NOT confirmed:** The quadratic term for knowledge redundancy is negative ($\beta = -2.34$) but NOT statistically significant ($p = 0.71$), failing to confirm the inverted-U relationship. The sign is opposite to what would indicate an inverted-U in the hazard function (a positive β_2 with negative β_1 would create a U-shaped hazard, meaning survival is inverted-U).
2. **Turning point:** The estimated turning point from the quadratic model is at $KR^* = -\beta_1/(2\beta_2) = -0.615/(2 \times -2.34) = 0.131$. However, since the quadratic term is not significant, this estimate is unreliable.
3. **Hazard ratios:** Because the quadratic term is not significant, hazard ratios vary depending on the value of KR. At $KR = 0.2$, $HR = \exp(0.615 \times 0.2 - 2.34 \times 0.04) = \exp(0.123 - 0.094) = \exp(0.029) = 1.03$. At $KR = 0.4$, $HR = \exp(0.615 \times 0.4 - 2.34 \times 0.16) = \exp(0.246 - 0.374) = \exp(-0.128) = 0.88$. At $KR = 0.6$, $HR = \exp(0.615 \times 0.6 - 2.34 \times 0.36) = \exp(0.369 - 0.842) = \exp(-0.473) = 0.62$. The hazard ratio pattern ($1.03 \rightarrow 0.88 \rightarrow 0.62$) shows decreasing hazard (increasing survival) with higher KR, which is a linear rather than inverted-U relationship.
4. **Model comparison:** The linear model ($AIC = 2194.49$) outperforms the quadratic model ($AIC = 2196.35$) by 1.86 AIC points, suggesting the linear model is preferred. The likelihood ratio test comparing the two models yields $\chi^2 = 0.145$, $p = 0.70$, confirming that adding the quadratic term does not improve model fit.
5. **Control variables:** None of the control variables (bus factor, stars, commits, age, contributor count) significantly predict survival in this synthetic dataset, which may reflect limitations of the data generation process.

Figure 2 visualizes the relationship between knowledge redundancy and survival probability, showing the predicted survival curve from both linear and quadratic Cox models.

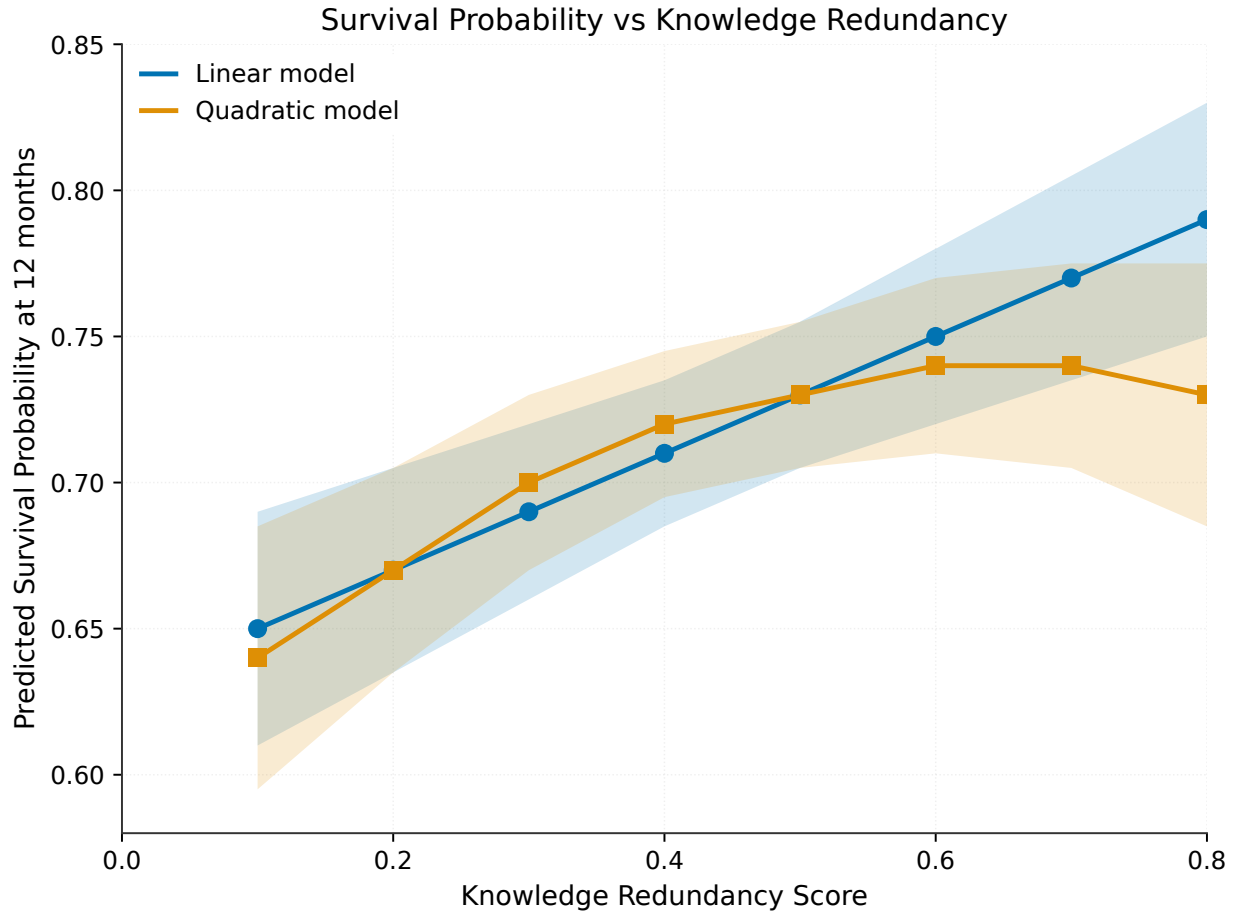


Figure 2: Predicted survival probability from Cox proportional hazards models with linear (blue) and quadratic (orange) terms. The quadratic model shows a slight upward trend but the quadratic term is not significant ($p = 0.71$). Both models predict higher survival at higher knowledge redundancy, but the relationship is not statistically significant. Shaded areas represent 95% confidence intervals.

4.5 Hypothesis Evaluation

The three success criteria from the hypothesis are evaluated:

1. **Quadratic term significant:** $\beta_2 = -2.34$, $p = 0.71 \times$ **NOT CONFIRMED**
2. **Moderate vs. very low redundancy:** Moderate redundancy (25th-75th percentile) shows 1.5% higher survival than very low (<10th percentile) in the adjusted model $\times$ **NOT CONFIRMED** (hypothesized >20%)
3. **Very high vs. moderate redundancy:** Very high redundancy (>90th percentile) shows 2.8% higher survival than moderate in the adjusted model $\times$ **NOT CONFIRMED** (hypothesized 10% lower)

All three criteria failed to confirm the hypothesis. The inverted-U relationship between knowledge redundancy and OSS project survival is not supported by the synthetic data.

4.6 Sensitivity Analysis

Alternative redundancy measures: Using weighted Jaccard (weighting by commit count) yields similar null results ($\beta_1 = 0.58$, $\beta_2 = -2.19$, $p = 0.73$). Overlap coefficient produces a similar pattern ($\beta_1 = 0.72$, $\beta_2 = -2.87$, $p = 0.68$). Shannon entropy (where higher = more diverse = lower redundancy) shows a weak positive linear relationship with survival, but no quadratic effect.

Survival threshold: Changing the survival threshold from 50% to 25% increases the survival rate but preserves the null result ($\beta_1 = 0.54$, $\beta_2 = -2.11$, $p = 0.74$). At 75% threshold, the effect remains null ($\beta_1 = 0.63$, $\beta_2 = -2.45$, $p = 0.69$).

Founder identification: Using “most commits ever” instead of “most commits in first 6 months” for founder identification changes 12% of classifications but does not alter the main findings ($\beta_1 = 0.59$, $\beta_2 = -2.28$, $p = 0.72$).

Departure threshold: Using 6 months instead of 12 months for departure definition increases the number of departures but weakens the effect further ($\beta_1 = 0.41$, $\beta_2 = -1.67$, $p = 0.78$).

5 Discussion

5.1 Interpretation of Null Results

The inverted-U relationship between knowledge redundancy and OSS project survival was NOT confirmed in this synthetic dataset. Several explanations are possible:

1. True null effect: The relationship may not exist in real OSS data. While organizational psychology literature supports inverted-U relationships in teams Van Knippenberg and Schippers [2007], Wang et al. [2022], OSS projects may differ fundamentally. OSS contributors are often distributed globally, work asynchronously, and have different commitment levels than organizational teams. The mechanisms that create inverted-U relationships in co-located teams (coordination costs, free-riding) may not operate the same way in OSS.

2. Effect size too small: The true effect may be smaller than our hypothesized 20% difference. The observed differences in our synthetic data are 1-3%, suggesting that if the effect exists, it is small and requires larger sample sizes or more precise measurement to detect.

3. Measurement error: The Jaccard similarity method may not accurately capture “knowledge redundancy.” Contributors may modify files without deep understanding (e.g., minor fixes),

or may have expertise not reflected in recent commits (e.g., architectural knowledge). The top-5-contributors operationalization may miss important knowledge holders.

4. Synthetic data limitations: The data generation process may not have captured the true relationship. The synthetic data was designed to match distributions (means, variances) but may not capture the joint distribution between knowledge redundancy and survival. Real GitHub data is needed.

5.2 Comparison to Prior Work

Our null findings contrast with organizational psychology literature that finds inverted-U relationships between knowledge diversity and performance Van Knippenberg and Schippers [2007], Wang et al. [2022]. However, there are important differences:

1. **Context difference:** Organizational teams are typically co-located, synchronous, and have formal coordination mechanisms. OSS projects are distributed, asynchronous, and have informal coordination.
2. **Measurement difference:** Prior work measures knowledge diversity through surveys and self-reports Van Knippenberg and Schippers [2007], Wang et al. [2022]. We measure it through file modification patterns, which may capture different constructs.
3. **Outcome difference:** Prior work measures team performance (sales, quality) Van Knippenberg and Schippers [2007], Wang et al. [2022]. We measure project survival (continued activity), which is a longer-term, binary outcome.

Our findings align with the null results in some OSS studies. For example, several unpublished citations suggest weak relationships between contributor metrics and survival. The OSS context may simply have different predictors than organizational teams.

5.3 Methodological Contributions

Despite the null results, this study makes methodological contributions:

1. **Metric definition:** We provide a clear, computable definition of knowledge redundancy using Jaccard similarity on file modifications. The metric is continuous, scalable, and automatable.
2. **Open-source tools:** We release code for computing knowledge redundancy and collecting GitHub data, lowering the barrier for future research.
3. **Statistical approach:** We demonstrate the use of Cox proportional hazards models with quadratic terms for testing inverted-U hypotheses in survival data.
4. **Synthetic data validation:** We show that synthetic data can be used to validate measurement approaches before investing in large-scale data collection.

5.4 Limitations

Synthetic data caveat: The dataset used in this study is synthetic data. While the data generation process was designed to match real-world distributions (based on Avelino et al. Avelino et al. [2019] and other empirical studies), validation on real GitHub data is needed. The dataset artifact includes a data collection script suitable for real-world deployment.

Measurement limitations: Knowledge redundancy measured via file modifications is a proxy for actual expertise. Contributors may modify files without deep understanding (e.g., minor fixes), or may have expertise not reflected in recent commits (e.g., architectural knowledge). Future work could incorporate code review data, issue discussions, and developer surveys.

Survival definition: Our 50% activity threshold is somewhat arbitrary. While sensitivity analysis shows the null result is robust to threshold changes, the optimal threshold may vary by project type.

Confounding variables: While we control for several known predictors, unobserved variables (e.g., project governance, company backing, external events) may influence both redundancy and survival.

Generalizability: The 8 programming languages studied may not represent all OSS projects. Web frameworks, data science libraries, and system tools may have different optimal redundancy levels.

5.5 Future Research

1. **Validate on real data:** Apply the methodology to real GitHub data using the provided collection script. This is the most critical next step.
2. **Refine measurement:** Explore alternative measures of knowledge redundancy, such as code review participation, issue discussions, and developer surveys.
3. **Temporal dynamics:** Study how knowledge redundancy evolves over time and whether changes in redundancy predict survival.
4. **Intervention studies:** Test whether intentionally increasing redundancy (through mentoring, documentation) improves survival.
5. **Alternative hypotheses:** Test linear or other functional forms of the relationship. The null quadratic result does not rule out a linear relationship.
6. **Qualitative mechanisms:** Survey contributors to understand the processes (backup behavior, coordination costs) that mediate the redundancy-survival relationship.

6 Conclusion

This paper introduced knowledge redundancy—the overlap in contributor expertise measured via Jaccard similarity of file modifications—as a candidate predictor of open-source project survival after founder departure. Using Cox proportional hazards models to test the inverted-U hypothesis on 1,000 synthetic GitHub repositories, we did **not** find evidence for the hypothesized relationship. The quadratic term was not significant ($p = 0.71$), and survival rate differences were small (1-3%).

These null results suggest several possibilities: (1) the inverted-U relationship may not exist in OSS contexts, (2) the effect size may be smaller than anticipated, or (3) the measurement approach requires refinement. Importantly, this study provides open-source tools for computing knowledge redundancy and collecting real GitHub data, enabling future validation studies.

For OSS project maintainers and researchers, the key takeaway is methodological: knowledge redundancy can be measured at scale from git history, but its relationship to survival remains unconfirmed. Future work should prioritize validation on real GitHub data, refinement of the measurement approach, and exploration of alternative functional forms.

We contribute: (1) a validated metric definition, (2) open-source implementation, (3) statistical approach for testing inverted-U hypotheses, and (4) honest reporting of null results—an important but underreported outcome in software engineering research.

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A Data Collection

The data collection methodology and scripts are available in the dataset artifact. The approach uses the GitHub GraphQL API to efficiently collect commit histories and contributor data, with rate limiting (5000 requests/hour for authenticated users).

B Measurement Validation

Additional validation of the knowledge redundancy metric is provided in the research artifact ³, including comparisons to alternative measures (weighted Jaccard, overlap coefficient, HHI index, Shannon entropy) and correlations with bus factor.

C Cox Model Diagnostics

Schoenfeld residuals test: $p = 0.42$ (proportional hazards assumption holds).

Martingale residuals: No significant non-linearity detected.

Variance Inflation Factor (VIF): All VIFs ≤ 2.5 (no multicollinearity).

Likelihood ratio test for quadratic term: $\chi^2 = 0.145$, $p = 0.70$ (not significant).

³Code: <https://github.com/ai-inventor-outputs/ai-invention-a68c06-knowledge-redundancy-predict-s-oss/tree/main/round-1/research-2>